

Solutions to JEE Main - 5 | JEE-2022

PHYSICS

SECTION-1

1.(A) $g = \frac{GM}{R^2} = \frac{G}{R^2} \left(\rho \left(\frac{4}{3} \pi R^3 \right) \right) = \frac{4}{3} G \pi \rho R$

2.(B) We know that the acceleration of the CM of the disc, $a_{CM} = \frac{g \sin \theta}{1 + \frac{k^2}{r^2}} = \frac{2}{3} g \sin \theta \left[k = \frac{r}{\sqrt{2}} \text{ for disc} \right]$

Therefore, $\frac{h}{\sin \theta} = \frac{1}{2} a_{CM} t^2$
 $\Rightarrow t = \frac{1}{\sin \theta} \sqrt{\frac{3h}{g}}$

3.(D) Let the relative density of the object be x and density of water be ρ

Then, buoyant force, $B = \rho v g = \rho \frac{m}{x \rho} g = \frac{mg}{x}$

In equilibrium, $T + mg = B$

$\Rightarrow x = \frac{mg}{mg + T}$

4.(B) We know that the orbital speed, $v = \sqrt{\frac{GM}{r}}$

So, the time period, $T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{GM}}$

Kinetic energy, $K = \frac{1}{2} m v^2 = \frac{GMm}{2r}$

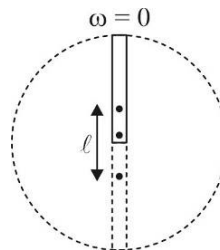
Since $K \propto r^{-1}$ and $T \propto r^{3/2}$

Therefore, $K \propto T^{-2/3}$

5.(D) To complete the loop, the angular speed of the rod when it reaches the top should be greater than zero. For $u_{\min}, \omega = 0$.

Applying energy conservation: $\left[\omega_0 = \frac{u}{\ell/2} = \frac{2u}{\ell} \right]$

$mg\ell = \frac{1}{2} \frac{m\ell^2}{3} \left(\frac{2u}{\ell} \right)^2 \Rightarrow u = \sqrt{\frac{3}{2} g\ell}$



6.(C) Time taken by the particle to reach the ground, $T = \sqrt{\frac{2h}{g}}$

Distance of the point of landing from the foot of the building, $R = vT = v \sqrt{\frac{2h}{g}}$

Let us take the point of projection as the origin, the +X direction towards right and the +Y direction upwards

Then, velocity of the particle just before landing, $\vec{v} = v\hat{i} - \sqrt{2gh}\hat{j}$

And position of the particle just before landing, $\vec{r} = v\sqrt{\frac{2h}{g}}\hat{i} - h\hat{j}$

Therefore, just before landing, the angular momentum of the particle about the point of projection,

$$\vec{L} = \vec{r} \times (m\vec{v}) = m \left(v\sqrt{\frac{2h}{g}}\hat{i} - h\hat{j} \right) \times (v\hat{i} - \sqrt{2gh}\hat{j}) = -mvh\hat{k}$$

7.(B) Equating the pressure at the bottom of both limbs,

$$P_0 + \rho_1 g(5) + \rho_2 g(4) = P_0 + \rho_2 g(6) \Rightarrow \frac{\rho_1}{\rho_2} = \frac{2}{5}$$

$$8.(A) AV = \text{Constant} \Rightarrow \frac{V_1}{V_2} = \frac{A_2}{A_1} = \frac{d_2^2}{d_1^2}; \frac{V_1}{V_2} = \frac{4^2}{2^2} = 4$$

9.(D) Since the only external force on the object – the force of gravity applied by the earth – is always directed towards the centre of the earth, the torque on the object about the centre is always zero. Hence, its angular momentum about the centre remains constant.

10.(B) For equilibrium, the net torque due to gravity on the body about the point O must be zero. Therefore, the CM of the body must be in the same vertical line as O. Taking O as origin:

$$X_{CM} = \frac{m_0 \left(-\frac{L}{2} \sin 30^\circ \right) + 2m(0) + m \left(\frac{L}{2} \right)}{m_0 + 2m + m}$$

We know that X_{CM} must be zero.

$$\text{Therefore, } m_0 = \frac{m}{\sin 30^\circ} \Rightarrow m_0 = 2m$$

11.(C) If the fan rotates with a constant angular velocity, then the net torque on it must be zero, i.e. the torque applied by the electric motor must be equal and opposite to the torque due to air resistance.

$$\tau_{motor} = \tau_{air} \Rightarrow \tau_{motor} \propto \omega^{3/2}$$

So, electric power, $P = \tau_{motor} \omega$, is proportional to $\omega^{5/2}$

$$12.(B) g = \frac{GM}{R^2}$$

$$g' = \frac{GM}{(R+h)^2} \approx \frac{GM}{R^2} \left[1 - \frac{2h}{R} \right] \quad (\text{because } h \ll R)$$

$$\text{For } h = \frac{R}{100}, \quad g' = \frac{GM}{R^2} \left(1 - \frac{2}{100} \right) = \left(\frac{98}{100} \right) g$$

13.(A) Velocity with which the liquid leaves the hole, $v = \sqrt{2gh}$

Time taken by any particle of the liquid to travel from the hole to the ground, $t = \sqrt{\frac{2(H-h)}{g}}$

$$\text{Therefore, } L = vt = (\sqrt{2gh}) \left(\sqrt{\frac{2(H-h)}{g}} \right) = 2\sqrt{h(H-h)}$$

14.(A) Taking rightward direction as positive,

$$F_A = \frac{G(m)(m)}{L^2} + \frac{G(m)(2m)}{(2L)^2} = \frac{3}{2} \frac{Gm^2}{L^2}$$

$$F_B = -\frac{G(m)(m)}{L^2} + \frac{G(m)(2m)}{L^2} = \frac{Gm^2}{L^2}$$

$$F_C = -\frac{G(2m)(m)}{(2L)^2} - \frac{G(2m)(m)}{L^2} = -\frac{5}{2} \frac{Gm^2}{L^2}$$

$$\text{So, } |F_A| : |F_B| : |F_C| = \frac{3}{2} : 1 : \frac{5}{2} = 3 : 2 : 5$$

15.(C) We know that

$$P_B = P_A + \frac{1}{2} \rho \omega^2 a^2$$

$$P_D = P_A + \rho g a$$

$$P_C = P_D + \frac{1}{2} \rho \omega^2 a^2 = P_A + \rho g a + \frac{1}{2} \rho \omega^2 a^2$$

Therefore, $P_C > P_A$ for all values of ω and $P_B > P_D$ only if $\omega > \sqrt{\frac{2g}{a}}$

16.(D) Let the density of the cube be ρ_o

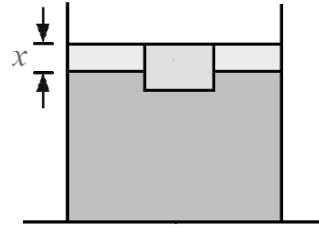
Then, because cube is in equilibrium $\rho_o a^3 g = \rho \left(\frac{a^3}{2} \right) g$

$$\Rightarrow \rho_o = \frac{\rho}{2}$$

Let the final equilibrium position of the cube be as shown

Then, $\rho_o a^3 g = \left(\frac{\rho}{3} \right) (a^2 x) g + \rho (a - x) (a^2) g$

$$\Rightarrow x = \frac{3}{4} a$$



17.(B) Conserving energy,

Loss is GPE = Gain in KE

$$mg(R \sin \theta) = \frac{1}{2} m (r\omega)^2 + \frac{1}{2} \left(\frac{2}{5} mr^2 \right) \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{10gR \sin \theta}{7r^2}}$$

18.(D) We already know that the escape velocity from the surface of the earth, $v_e = \sqrt{\frac{2GM}{R}}$

Conserving energy:

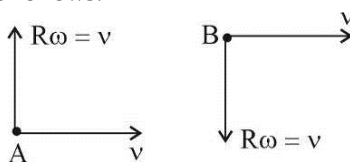
$$U_i + K_i = U_f + K_f$$

$$-\frac{GMm}{R} + \frac{1}{2} m \left(\frac{v_e}{\sqrt{3}} \right)^2 = -\frac{GMm}{(R+h)}$$

Solving, we get

$$h = \frac{R}{2}$$

- 19.(A) Let us assume the disc is moving towards right. Then, the translational and rotational components of the velocities of A and B are as follows:



Therefore, taking the +X direction towards the right and the +Y direction upwards,

$$\vec{v}_A = v\hat{i} + v\hat{j}$$

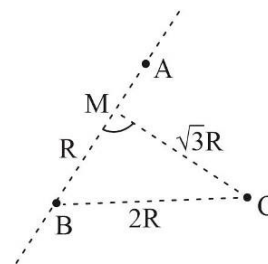
$$\vec{v}_B = v\hat{i} - v\hat{j}$$

So, $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 2v\hat{j}$

- 20.(B) The axis of rotation passes through the centres of two of the spheres, A and B, and the axis is at a distance $\sqrt{3}R$ from the centre of the third sphere, C.

So, using parallel axis theorem for the third sphere, the total moment of inertia,

$$I = 2 \times \left(\frac{2}{5} mR^2 \right) + \left(\frac{2}{5} mR^2 + m(\sqrt{3}R)^2 \right) = \frac{21}{5} mR^2$$



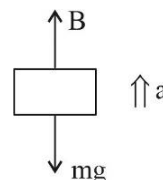
SECTION-2

- 21.(15) Let the volume of the block be V and let its acceleration just after it is released be a

$$B - mg = ma$$

$$\Rightarrow V(10^3)g - V(400)g = V(400)a$$

$$\Rightarrow a = \frac{3}{2}g = 15m/s^2$$



- 22.(16) We do not need the exact formula for the moment of inertia of a triangular plate about one of its sides to find the answer to this question.

If the side length of the triangle is a and the mass of the triangle is m , then the moment of inertia will be proportional to ma^2 .

Now, the mass m will itself be proportional to a^2 (because the mass will be proportional to the area of the triangle)

So, the moment of inertia is proportional to a^4

- 23.(40) Let the fraction of Metal A be x

Let the total volume of the sphere be V

Then, Weight = Buoyant force

$$\Rightarrow [8x + 6(1-x)]Vg = \frac{1}{2}(13.6)Vg$$

$$\Rightarrow x = 0.4$$

- 24.(2) Let atmospheric pressure be P_o . Suppose the pressure inside the tube is $6P_o$ (the maximum allowed) and the liquid comes out of the hole at speed v .

Since the hole is very small, the velocity of the liquid inside the tube will be negligible

Applying Bernoulli's theorem,

$$6P_o = P_o + \frac{1}{2}\rho v^2$$

$$\Rightarrow v = \sqrt{\frac{10P_o}{\rho}} = \sqrt{\frac{10 \times 10^5}{10^4}} = 10 \text{ m/s}$$

Therefore, $Q_{\max} = v \times (\text{area of cross-section of the hole})$

$$\Rightarrow Q_{\max} = (10)(2 \times 10^{-6}) = 2 \times 10^{-5} \text{ m}^3/\text{s}$$

- 25.(3) At this instant, the velocity of the particle, $v_1 = 2R\omega$

So, kinetic energy of the particle,

$$K_1 = \frac{1}{2}m(2R\omega)^2 = 2mR^2\omega^2$$

Kinetic energy of the ring,

$$K_2 = \frac{1}{2}m(R\omega)^2 + \frac{1}{2}(mR^2)\omega^2 = mR^2\omega^2$$

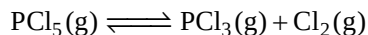
So, total kinetic energy,

$$K = K_1 + K_2 = 3mR^2\omega^2$$

Chemistry

SECTION-1

- 1.(D) Expansion of the Universe is a spontaneous process such that $\Delta S_{\text{sys}} > 0$; $\Delta S_{\text{surr}} = 0$ and $\Delta G_{\text{sys}} < 0$
- 2.(B) By adding an inert gas at constant pressure; yield of the products can increase only for completely reversible gaseous reactions having $\Delta n_{(g)} > 0$. It is valid only for the option (B).



- 3.(C) At 298 K and 1 atm ; $\Delta_f H_{\text{H}_2\text{O}(\ell)}^\circ \neq 0$, $\Delta_f H_{\text{H}(\text{g})}^\circ \neq 0$ but $\Delta_f H_{\text{H}^+(\text{aq})}^\circ = 0$ (according to the IUPAC convention). At zero Kelvin some substances have some residual entropy.

- 4.(A) For the given equilibrium $\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{H}_2\text{O}(\ell)$

By increasing pressure over the system melting point of ice will decrease and amount of water will increase.

- 5.(B) During adiabatic process there will be NO heat exchange between the system and surrounding.

Also $\Delta U = w$,

So adiabatic expansion involve cooling of the gas

As $|W|_{\text{rev}} > |W|_{\text{irrev}}$ for expansion of the gas thus more cooling of the gas will occur during reversible process.

Adiabatic compression involve heating of the gas. As $|W|_{\text{rev}} < |W|_{\text{irrev}}$ for compression of the gas thus more heating of the gas will occur during irreversible process.

- 6.(B) $\Delta H = \Delta U + (\Delta n)_g RT$, where ΔH is heat exchange at constant pressure while ΔU is heat exchange at constant volume. Magnitude of ΔH will be lower than that of ΔU if $(\Delta n)_g$ is negative. It is applicable only for option (B)

- 7.(A) As $K_p = K_c(RT)^{\Delta n}$ for the reaction $2\text{NOCl}(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{Cl}_2(\text{g})$

$$\Delta n = +1, \text{ thus } K_p = K_c \times RT \quad \therefore \quad K_c = \frac{K_p}{RT} = \frac{3.5 \times 10^{-3}}{0.0821 \times 300} = 1.42 \times 10^{-4}$$

- 8.(C) (A) Chemical equilibrium inside a closed vessel is dynamic in nature.
 (B) After some time, radioactivity will be observed in both solution phase and solid phase due to dynamic nature of the equilibrium
 (C) Equilibrium $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{aq})$ is effected by change in temperature and pressure
 (D) Statement is not true for all reversible reactions

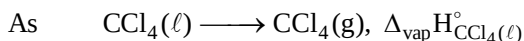
- 9.(D) $\text{CCl}_4(\text{g}) \longrightarrow \text{C}(\text{g}) + 4\text{Cl}(\text{g})$ (Reaction-1)

$$\Delta_r H = \Delta_f H_{\text{C}(\text{g})}^\circ + 4\Delta_f H_{\text{Cl}(\text{g})}^\circ - \Delta_f H_{\text{CCl}_4(\text{g})}^\circ \quad \text{Equation-1}$$

$$\Delta_f H_{\text{C}(\text{g})}^\circ = \text{standard molar enthalpy of atomization of graphite} = 720 \text{ kJ/mole}$$

$$\Delta_f H_{\text{Cl}(\text{g})}^\circ = \frac{1}{2} (\text{standard molar enthalpy of atomization of } \text{Cl}_2(\text{g}))$$

$$= \frac{1}{2}(240) = 120 \text{ kJ / mole}$$



$$\Delta_{\text{vap}}H_{\text{CCl}_4(\ell)}^\circ = \Delta_f H_{\text{CCl}_4(\text{g})}^\circ - \Delta_f H_{\text{CCl}_4(\ell)}^\circ \quad \text{Equation-2}$$

$$30 = \Delta_f H_{\text{CCl}_4(\text{g})}^\circ - [-130]$$

$$\Delta_f H_{\text{CCl}_4(\text{g})}^\circ = -100 \text{ kJ / mole}$$

$\therefore$ For reaction-1

$$\Delta_r H = 720 + 4[120] - [-100] = 720 + 480 + 100 = 720 + 580 = 1300 \text{ kJ}$$

10.(A) For $(P_1, V_1, T_1) \longrightarrow (P_2, V_2, T_2)$ thermodynamic process magnitude of work done by the ideal gas system is in order.

Isobaric > Isothermal > Adiabatic > Isochoric
(option A) (option B) (option D) (option C)

11.(A) As the process is isothermal for an ideal gas thus $\Delta U = 0, Q \neq 0, W \neq 0$. (as $P_{\text{ext}} \neq 0$), $\Delta S \neq 0$

12.(B) $p \rightarrow q$ (Isochoric heating)

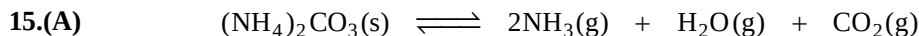
$q \rightarrow r$ (Isobaric heating)

$r \rightarrow s$ (Isochoric cooling)

$s \rightarrow p$ (Isobaric cooling)

13.(B) For $\text{N}_2\text{O}_4 \rightleftharpoons \text{N}_2 + 2\text{O}_2$ $K_p = \frac{1}{5 \times 10^{-2} \times 10^{-1}} = \frac{1000}{5} = 200 \text{ atm}$

14.(C) Decomposition of PCl_5 is an endothermic reaction and for such reaction yield of the product increases with temperature.



Initially 0.2 atm
At equil. $(0.2 + 2P) \text{ atm} \quad P \text{ atm} \quad P \text{ atm}$

$\therefore$ Total pressure of the gases at equilibrium $= (0.2 + 4P) = 1$

$$\Rightarrow 4P = 0.8 \text{ or } P = 0.2 \text{ atm}$$

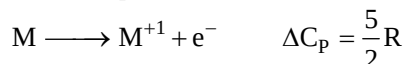
As $K_p = (0.2 + 2P)^2 (P)(P) = (0.36)(0.04) = 1.44 \times 10^{-2}$

16.(C) In condensation, freezing and crystallization, entropy decreases while vaporization is endothermic and entropy is increases hence it is entropy driven.

17.(D) For processes in which one of the variable $[P, T \text{ or } V]$ is constant, the $[\Delta S_{\text{sys}}]_{\text{rev}} = [\Delta S_{\text{sys}}]_{\text{irrev}}$

As during an adiabatic process neither of these variable $[P, T \text{ or } V]$ is fixed thus $[\Delta S_{\text{sys}}]_{\text{rev}} \neq [\Delta S_{\text{sys}}]_{\text{irrev}}$

18.(C) For the ionization process



As $\Delta_r H_{(T)} = \Delta_r H_{(O)} + \Delta C_p T$

$(\Delta_r H_{(T)}) = \text{ionisation enthalpy at temperature}$

$(\Delta_r H_{(O)}) = \text{ionisation energy at absolute zero}$

$$\therefore \Delta_r H_{(T)} = E_0 + \frac{5}{2} RT$$

$$19.(D) \quad |W|_{\text{Adb}} = C_V[\Delta T] = \frac{R}{(\gamma-1)}[\Delta T]$$

$$\text{Here } \gamma = \frac{C_P}{C_V} = \left(1 + \frac{2}{f}\right), \quad f = \text{degree of freedom}$$

As degree of freedom (f) increases with atomicity of the gas molecules thus magnitude of work done increases with atomicity of the gas.

20.(A) Reaction with maximum value of K_c will be nearest to completion.

The reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightleftharpoons 2\text{HBr}$ have maximum value of K_c at the given temperature.

SECTION-2

21.(41.40) Average molecular mass of a mixture of NO_2 and N_2O_4 having 80% of the dimer is

$$\frac{46 \times 20 + 92 \times 80}{100} = 82.8$$

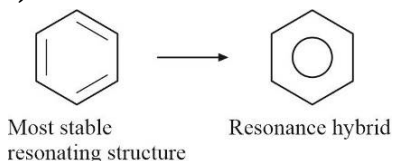
$$\text{Vapour density} = \frac{M_{\text{avg}}^{\circ}}{2} = 41.4$$

$$22.(1.38) \quad \Delta S = nC_V \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$\text{Here } n = 2 \quad C_V = \frac{3}{2}R \quad (\text{for monoatomic gas})$$

$$\begin{aligned} \Delta S &= 2 \left[\frac{3}{2}R \right] \ln 2 + 2R \ln \left(\frac{1}{2} \right) = 3R \ln 2 + 2R[-\ln 2] = R \ln 2 \\ &= 2 \ln 2 = 2 \times 0.69 = 1.38 \text{ cal/K} \end{aligned}$$

23.(-149)



$$\Delta_r H = \text{Resonance energy of benzene} = 3[-119] - [-208] = -357 + [208] = -149 \text{ kJ/mole}$$

24.(1) For the reaction $\text{AgCl} + 2\text{NH}_3 \rightleftharpoons \text{Ag}(\text{NH}_3)_2\text{Cl}(\text{aq})$

$$\Delta_r G^{\circ} = \Delta_f G_{\text{Ag}(\text{NH}_3)_2\text{Cl}}^{\circ} - [\Delta_f G_{\text{AgCl}}^{\circ} + 2\Delta_f G_{\text{NH}_3}^{\circ}] = -40 - [-20 + 2(-10)] = 0$$

$$\text{As } \Delta_r G^{\circ} = -2.303RT \log_{10} K \Rightarrow \log_{10} K = 0 \text{ or } K = 1$$

$$25.(31.70) \quad C_P = \frac{Q_P}{n \Delta T} = \frac{300}{3 \times 2.5} = 40 \text{ J/mole-K}$$

$$\text{As } C_P - C_V = R \quad (\text{for ideal gases})$$

$$\Rightarrow C_V = C_P - R = 40 - 8.3 = 31.7 \text{ J/mole-K}$$

Mathematics

SECTION-1

1.(B) $S_1 = 4 + 9 - 4 + 12 - 4 = 17$

2.(C) Here $C_1 \equiv (2, -3), C_2 \equiv (2, -2) r_1 = \lambda \quad r_2 = 3$

Hence $C_1 C_2 < r_1 + r_2$

$\Rightarrow 1 < \lambda + 3 \Rightarrow \lambda > -2 \quad \dots(i)$

and $|r_1 - r_2| < C_1 C_2 \Rightarrow |\lambda - 3| < 1 \Rightarrow 2 < \lambda < 4$

Using (i) and (ii) $2 < \lambda < 4$

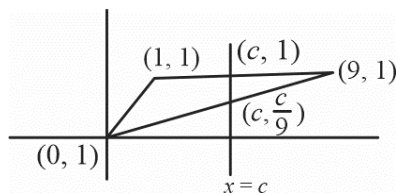
3.(A) If ordinate is -1 then abscissa may be 3 or 5. Hence normal are lines passing through $(4, 1)$ and $(3, -1)$ and other passes through $(4, 1)$ $(5, -1)$

4.(B) Area of $\Delta = \frac{1}{2} \times 8 \times 1 = 4$

Area of small $\Delta^{le} = \frac{1}{2}(9-c) \times \left(1 - \frac{c}{9}\right) = 2$

$(a-c)^2 = 36 \Rightarrow 9-c=6 \text{ or } -6$

$c=3$



5.(A) $\left(a^{1/13} + a^{3/2}\right)^n$

$T_2 = {}^nC_1 \left(a^{1/13}\right)^{n-1} a^{(3/2)}$

$= {}^nC_1 a^{\left(\frac{n-1}{13} + \frac{3}{2}\right)} = 14a^{5/2}$

$\Rightarrow n=14$

$\frac{{}^{14}C_3}{{}^{14}C_2} = \frac{\frac{14 \times 3 \times 12}{6}}{\frac{14 \times 13}{2}} = 4$

6.(B) ${}^{39}C_{3r-1} + {}^{39}C_{3r} = {}^{39}C_r + {}^{39}C_{r^2-1}$

${}^{40}C_{3r} = {}^{40}C_{r^2}$

$3r = r^2 \quad \text{or} \quad r^2 + 3r = 40$

$r=0, r=3 \quad \text{or} \quad r^2 + 3r - 40 = 0 \Rightarrow r = -8, r=5$

$r=5$

Possible value $\{3, 5\}$

7.(A) Intersection of both lines is $(2, -5)$ is centre and $2\pi r = 8\pi \Rightarrow r=4$

Hence circle $\Rightarrow (x-2)^2 + (y+5)^2 = 4^2$

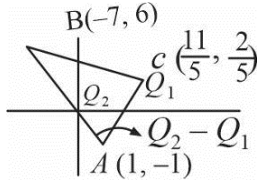
- 8.(D) Here centre of circle $(2, -3)$ is mid point of both the ends of diameter hence

$$\begin{array}{ccc} \bullet & \bullet & \bullet \\ (x, y) & (2, -3) & (6, 0) \end{array}$$

$$\Rightarrow \frac{x+6}{2} = 2 \Rightarrow x = -2$$

$$\text{And } \frac{y+0}{2} = -3 \Rightarrow y = -6$$

9.(B) $\tan(\theta_2 - \theta_1) = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{\frac{8}{5} - 2}{1 - \frac{16}{6}}$



$$\tan \theta = \frac{20}{10} = 2$$

10.(B) $(5-1)^{87} + (5+1)^{87}$
 $= 2 \left[{}^{87}C_0 5^{87} + {}^{87}C_2 5^{85} + {}^{87}C_4 5^{83} \dots + {}^{87}C_{86} 5^1 \right]$

$$\text{Remainder when divided by } 25 = \frac{87 \times 10}{25} = \frac{870}{25}$$

$$\text{Remainder} = 20$$

11.(A) $\sum_{k=0}^4 \frac{4! x^k}{k!(4-k)!} \cdot 5^{4-k} = \frac{(x+5)^4}{4!} = \frac{8}{3}$

$$(x+5)^4 = 64$$

$$x+5 = 2\sqrt{2} \text{ or } -2\sqrt{2}$$

$$x = 2\sqrt{2} - 5, \quad x = -2\sqrt{2} - 5$$

12.(A) $\tan(45^\circ) = \left| \frac{\frac{4a+3}{a+1} - \frac{3}{4}}{1 + \frac{12a+9}{4a+4}} \right| = \left(\frac{16a+12-3a-3}{16a+13} \right)$

$$\frac{13a+9}{16a+13} = 1$$

$$13a+9 = 16a+13 \text{ or } 29a = -12$$

$$3a = -4 \Rightarrow a = \frac{-4}{3} \quad a = \frac{-12}{29}$$

- 13.(B) Direct condition of normality.

- 14.(B) According to given conditions, only internal touch is possible

$$\text{Hence } C_1 C_2 = |r_1 - r_2| \Rightarrow |c| = 2$$

15.(A) Put $x=1$ $a_0 + a_1 + a_2 + a_3 + \dots + a_{50} = 3^{25}$

Put $x=-1$ $\frac{a_0 - a_1 + a_2 - a_3 + \dots + a_{50}}{2(a_0 + a_2 + a_4 + \dots)} = \frac{1}{3^{25} + 1}$

$$a_0 + a_2 + a_4 + \dots = \frac{3^{25} + 1}{2}$$

$$= \frac{(4-1)^{25} + 1}{2} = \text{even}$$

16.(C) $(1+x)^5 (3+x)^4 (7+x)^3$

Coefficient of $x^{11} = {}^3C_2 \times 7 + {}^4C_3 \times 3 + {}^5C_4$

$$= 21 + 12 + 5 = 38$$

17.(A) $(1-2x+5x^2-10x^3)(1+x)^n = 1 + a_1x + a_2x^2 + \dots$

$$a_1 = \text{coeff of } x = {}^nC_1 - 2 = n - 2$$

$$a_2 = \text{coeff of } x^2 = {}^nC_2 - 2 {}^nC_1 + 5$$

$$n^2 - 4n + 4 = 2 \left(\frac{n^2 - n}{2} - 2n + 5 \right)$$

$$n = 6$$

18.(A) $x^2 + y^2 + 2ax + c = 0$, $x^2 + y^2 + 2by + c = 0$

Common tangent at common point $ax - by = 0$

Perpendicular distance from centre = radius

$$x^2 + y^2 + 2ax + c = 0 \quad (-a, 0)$$

$$\left| \frac{-a^2}{\sqrt{a^2 + b^2}} \right| = \sqrt{a^2 - c} \Rightarrow a^4 = (a^2 + b^2)(a^2 - c)$$

$$a^4 = a^4 + a^2b^2 - c(a^2 + b^2)$$

$$\frac{1}{c} = \frac{1}{a^2} + \frac{1}{b^2}$$

19.(C) $\angle ACB = 90^\circ$, S is point of intersection of angle bisector

$$\text{hence } \angle SAB + \angle SBA = \frac{90^\circ}{2} = 45^\circ \Rightarrow \angle ASB = 135^\circ \text{ for all } C$$

20.(B) Putting given points in equation of circle, point $(-2, 1)$ gives negative value.

SECTION-2

21.(201) $2^{100} - \frac{\sum_{k=0}^{100} C_{k+1}}{101} = 2^{100} - \frac{[2^{101} - 1]}{101}$

$$= \frac{101 \cdot 2^{100} - 2^{101} + 1}{101}$$

$$= \frac{2^{100}(99) + 1}{101}$$

$$a + b + c = 201$$

22.(2) Here $C_1 C_2 = \sqrt{45}$ and $r_1 + r_2 = 8$ and $|r_1 - r_2| = 2$

$$\Rightarrow (r_1 - r_2) < C_1 C_2 < r_1 + r_2$$

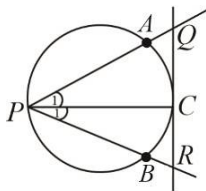
$\Rightarrow$ both circles are intersecting each other

$\Rightarrow$ two common tangents are possible

23.(5050) Highest exponent of $x = 1 + 2 + 3 + \dots + 100 = 5050$

24.(11) $QC^2 = QP \cdot QA$

$$9 = QP \cdot 2$$



$$QP = \frac{9}{2}$$

$$\frac{PQ}{PR} = \frac{QC}{RC} = \frac{3}{4}$$

$$PR = \frac{4PQ}{3} = \frac{4}{3} \times \frac{9}{2} = 6$$

$$(RC)^2 = RB \cdot RP$$

$$16 = RB \cdot 6$$

$$RB = \frac{8}{3}$$

25.(4) Co-ordinate of B will be point of intersection of $7x - 10y + 1 = 0$, $3x - 2y + 5 = 0$

$(-3, -2)$

Image of A in angle bisector of B lies on line BC .

